

Mohr's Circle for Plane stress

The transformation equations for plane stress are given as,

$$\begin{aligned}\sigma_{x'x'} &= \left(\frac{\sigma_{xx} + \sigma_{yy}}{2}\right) + \left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right) \cos 2\theta + \tau_{xy} \sin 2\theta && \tau_{x'y'} = -\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right) \sin 2\theta + \tau_{xy} \cos 2\theta \\ \sigma_{y'y'} &= \left(\frac{\sigma_{xx} + \sigma_{yy}}{2}\right) - \left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right) \cos 2\theta + \tau_{xy} \sin 2\theta && \tau_{x'y'} = -\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right) \sin 2\theta + \tau_{xy} \cos 2\theta\end{aligned}$$

Squaring both sides of the equations

$$\left\{\sigma_{x'x'} - \left(\frac{\sigma_{xx} + \sigma_{yy}}{2}\right)\right\}^2 = \left\{\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right) \cos 2\theta + \tau_{xy} \sin 2\theta\right\}^2 \ll \ll \left\{\tau_{x'y'} - 0\right\}^2 = \left\{-\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right) \sin 2\theta + \tau_{xy} \cos 2\theta\right\}^2$$

Adding these two expressions,

$$\begin{aligned}\left\{\sigma_{x'x'} - \left(\frac{\sigma_{xx} + \sigma_{yy}}{2}\right)\right\}^2 + \left\{\tau_{x'y'} - 0\right\}^2 &= \left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right)^2 \cos^2 2\theta + \tau_{xy}^2 \sin^2 2\theta + 2\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right) \tau_{xy} \cos 2\theta \sin 2\theta \\ &\quad + \left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right)^2 \sin^2 2\theta + \tau_{xy}^2 \cos^2 2\theta - 2\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right) \tau_{xy} \cos 2\theta \sin 2\theta \\ \left\{\sigma_{x'x'} - \left(\frac{\sigma_{xx} + \sigma_{yy}}{2}\right)\right\}^2 + \left\{\tau_{x'y'} - 0\right\}^2 &= \left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right)^2 + \tau_{xy}^2\end{aligned}$$

$$\left\{\sigma_{x'x'} - \left(\frac{\sigma_{xx} + \sigma_{yy}}{2}\right)\right\}^2 + \left\{\tau_{x'y'} - 0\right\}^2 = \left\{\sqrt{\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right)^2 + \tau_{xy}^2}\right\}^2$$

This is now of the form $(x - h)^2 + (y - k)^2 = R^2$, which is the equation of a circle whose centre is at (h, k) and radius R .

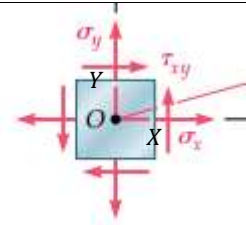
The transformation equations are parametric equations of a circle with parameter 2θ . This is known as **Mohr's circle** of plane stress. The centre of the

circle is at $\left(\frac{\sigma_{xx} + \sigma_{yy}}{2}, 0\right)$ and radius $= \sqrt{\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right)^2 + \tau_{xy}^2}$ and every angle θ will appear as 2θ in the Mohr's circle.

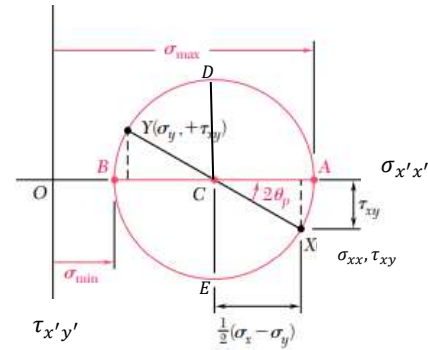
Every point on the Mohr's circle represents a plane on the body and the co-ordinates of the point represents the normal and shear stress acting on the plane.

Procedure:

1. $\sigma_{x'x'}$ is taken positive towards right.
2. $\tau_{x'y'}$ is taken positive downwards.
3. θ is taken positive counter-clockwise. Since the equation of Mohr circle is having parameter 2θ , every angle in the original system will appear as twice its value in Mohr circle.



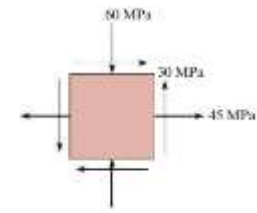
Let us consider the plane stress situation as shown. The plane whose normal is in the positive x-direction, is marked as X and the plane whose normal is in the positive y-direction, is marked as Y. The co-ordinates of $X(\sigma_{xx}, -\tau_{xy})$ are marked as per the convention. The plane Y is at 90° counter clockwise from plane X (their outward normals are at 90°). Hence, the plane Y will be at 180° counter clockwise from the plane X in the Mohr's circle with coordinates $Y(\sigma_{yy}, \tau_{xy})$.



That is, X and Y are diametrically opposite in the Mohr's circle. Join X and Y. It will intersect the x-axis at the centre C of the Mohr circle. With C as the centre and CX or CY as radius, draw a circle. This is the Mohr circle. The circle will intersect the x-axis at A and B. Since the y-coordinate of point A and B are zeroes (shear stress=0), these points represent the principal planes which are mutually perpendicular to each other. The $\angle XCA$ gives $2\theta_p$. The Maximum shear stress planes are at 45° from the principal planes. These planes will appear at 90° from the principal planes (A and B) in the Mohr circle. Hence, D and E are the planes of maximum shear stress. The coordinates of D are $\left(\frac{\sigma_{xx} + \sigma_{yy}}{2}, \sqrt{\left(\frac{\sigma_{xx} - \sigma_{yy}}{2} \right)^2 + \tau_{xy}^2} \right)$

#Example 1

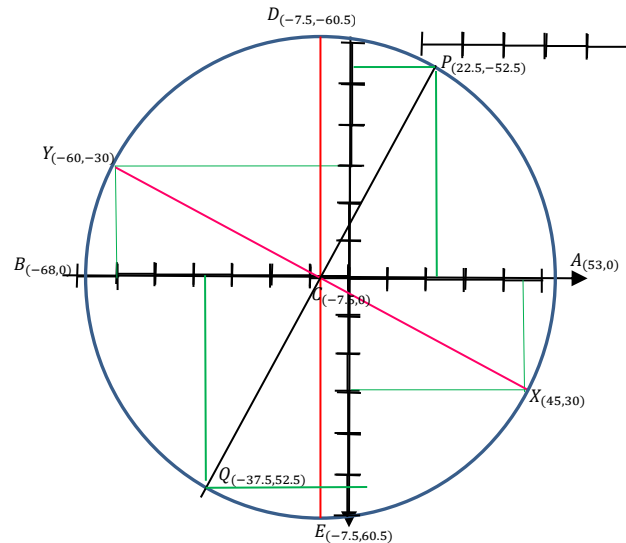
The state of stress at a point in a material is shown. Using Mohr's circle, determine the stresses on a plane whose normal makes an angle 45° with the x-axis. Also determine the principal stresses and maximum shear stress.



First choose a suitable scale ($1 \text{ cm} = 10 \text{ MPa}$).

Locate the face X with coordinates $(45, 30)$ and face Y with coordinates $(-60, -30)$. Join X and Y . It will intersect the x-axis at $C(-7.5, 0)$. With C as centre and radius CX or CY complete the circle. The circle will intersect the x-axis at $A(53, 0)$ and $B(-68, 0)$. The principal stresses are $\sigma_{11} = 53 \text{ MPa}$ and $\sigma_{22} = -68 \text{ MPa}$. Measure $\angle XCA$ which is equal to $2\theta_p$

$$2\theta_p \cong 30^\circ \ll \gg \theta_p = 15^\circ, 105^\circ$$



The principal planes are inclined at 15° and 105° to the x-axis.

The maximum shear stress planes are inclined at 45° to the principal planes. Locate the points D and E on the circle which are located at 90° from A and B . The coordinates of these points are $(-7.5, \pm 60.5)$. The maximum shear stresses are $\pm 60.5 \text{ MPa}$ and the normal stress on the planes is $\sigma_{ave} = -7.5 \text{ MPa}$.

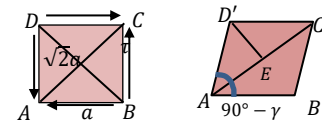
To find the stress on a plane whose normal is at 45° , locate a point P on the circle which is at 90° from X . The coordinates of P are $(22.5 \text{ MPa}, -52.5 \text{ MPa})$. The normal stress is 22.5 MPa , and shear stress is -52.5 MPa . The other perpendicular plane Q is having coordinates $(-37.5 \text{ MPa}, 52.5 \text{ MPa})$.

Exercise

Construct the Mohr circle for the plane state of stress shown below and determine the principal stresses, orientation of principal planes, maximum shear stress, and orientation of maximum shear stress planes. Also determine the normal and shear stress values on a plane whose normal is making 25° with x-axis.

Relation connecting E , G , and ν

Consider a square element with side ' a ' with a state of pure shear. The principal stresses are $+\tau$ along AC (at 45°) and $-\tau$ along BD (at 135°).



Length of line $AC = \sqrt{2}a$ $\ll$ The strain along the line AC , $\epsilon = \frac{\tau}{E} - \nu \left(-\frac{\tau}{E} \right) = \frac{\tau}{E} (1 + \nu)$

The change in length of AC , $\Delta_{ac} = \frac{\tau}{E} (1 + \nu) \sqrt{2}a$

If γ is the shear strain, $\gamma = \frac{\tau}{G}$ $\ll$ $\angle BAD' = 90^\circ - \gamma$

Draw a line $D'E \perp AC'$, then $AE = EC' = \frac{AC'}{2} = \frac{\sqrt{2}a + \Delta_{ac}}{2}$ $\ll$ $\angle EAD' = \frac{90^\circ - \gamma}{2} = 45^\circ - \frac{\gamma}{2}$

$$\cos \left(45^\circ - \frac{\gamma}{2} \right) = \frac{AE}{a} = \frac{\sqrt{2}a + \Delta_{ac}}{2a} = \frac{1}{\sqrt{2}} + \frac{\Delta_{ac}}{2a}$$

$$\cos 45 \cos \frac{\gamma}{2} + \sin 45 \sin \frac{\gamma}{2} = \frac{1}{\sqrt{2}} + \frac{\Delta_{ac}}{2a} \quad \ll \quad \frac{1}{\sqrt{2}} \cos \frac{\gamma}{2} + \frac{1}{\sqrt{2}} \sin \frac{\gamma}{2} = \frac{1}{\sqrt{2}} + \frac{\Delta_{ac}}{2a}$$

Since the shear strain γ is very small, $\cos \frac{\gamma}{2} \cong 1$ and $\sin \frac{\gamma}{2} \cong \frac{\gamma}{2}$

$$\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \frac{\gamma}{2} = \frac{1}{\sqrt{2}} + \frac{\Delta_{ac}}{2a} \quad \ll \quad \Delta_{ac} = \frac{a\gamma}{\sqrt{2}} = \frac{a}{\sqrt{2}} \frac{\tau}{G}$$

Comparing the two expressions for Δ_{ac} ,

$$\frac{a}{\sqrt{2}} \frac{\tau}{G} = \frac{\tau}{E} (1 + \nu) \sqrt{2}a \quad \ll \quad G = \frac{E}{2(1 + \nu)}$$

References:

1. *Mechanics of Materials*, 2/e, J. M. Gere & S. P. Timoshenko, CBS Publishers.
2. *Mechanics of Materials*, 6/e, F.P. Beer & E. R. Johnston, McGraw Hill
3. *Engineering Mechanics of Solids*, 2/e, E. P. Popov, Pearson Education Asia.
4. *Mechanics of Materials*, 9/e, R. C. Hibbeler, Prentice Hall.